Our Work in Welding

Advanced AI Solutions for Manufacturing Excellence

AI-Driven Welding Solutions

Visual Content

KEY POINTS

- • Implementation of AI technology to enhance welding quality and efficiency.
- • Real-time monitoring of welding processes through advanced data analytics.
- • Utilization of sound analysis models to detect and classify welding defects.

Data Collection for Quality Assurance

KEY POINTS

- ▸ • Systematic collection of welding data to log accurate sensor parameters.
- ▸ • Integration of multiple modalities: audio, video, and time-series data.
- ▸ • Automated data recording to ensure synchronization across various sensors.

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Experimental Setup and Configuration

KEY POINTS

- • Camera mounted on the welding robot for visual monitoring.
- • Microphones strategically placed to capture sound during the welding process.
- • Detailed setup ensures comprehensive data collection for analysis.

Dataset Requirements for Analysis

KEY POINTS

- ▸ • Dataset includes data from 4 modalities: audio, video, welding parameters, and photographs.
- ▸ • Minimum of 20 samples per defect type across 14 combinations of welding parameters.
- ▸ • Data must be time-synchronized for effective cross-analysis of welding activities.

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Classification and Documentation of Defects

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KEY POINTS

- ▸ • Classification of welds into 'Good' and 'Defective' categories.
- ▸ • Natural language descriptions provided for each weld, detailing the creation process.
- ▸ • Supporting testing data included to confirm the presence of defects.

Business Value and Future Directions

KEY POINTS

- • Enhanced quality control leads to reduced rework and increased customer satisfaction.
- • Data-driven insights facilitate continuous improvement in welding processes.
- • Future projects will expand on AI capabilities to further automate and optimize welding operations.

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